

ROI Analyses Hippocampus - Model 2 (Correct and Incorrect associations)

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Hippocampus")
ROIdata <- read.csv("BOLD_Hipp_model2.csv")

# Assign factors
ROIdata$S_ID <- factor(ROIdata$S_ID)
ROIdata$ID <- factor(ROIdata$ID)
ROIdata$ROI <- factor(ROIdata$ROI)
ROIdata$hemisphere <- factor(ROIdata$hemisphere)
ROIdata$StimType <- factor(ROIdata$StimType)
ROIdata$TaskStage <- factor(ROIdata$TaskStage)
ROIdata$Response <- factor(ROIdata$Response)
```

Sham Condition - segmented Hippocampus - Table S8 and Figure 2H

```
# LMM model comparing correct vs incorrect estimates for segmented hippocampus during the Sham condition
# BOLD = median % BOLD signal

## -- left hippocampus

Hip_data_seg_enc <- subset(ROIdata, StimType=="Sham" & (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & Response=="Correct")

Hipp_seg_Sham <- lmer(BOLD ~ Response*ROI + (1|ID), data = Hip_data_seg_enc)
Anova(Hipp_seg_Sham, test="F")

## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##
```

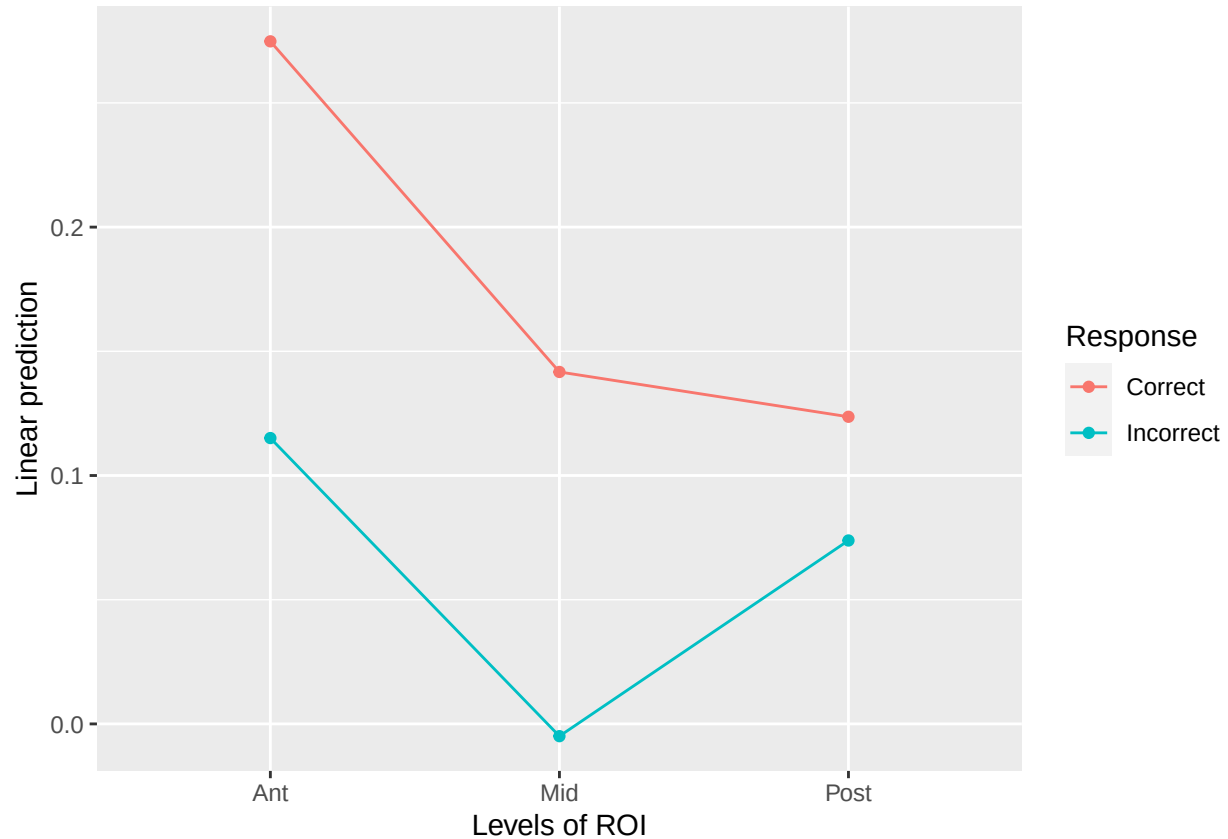
	F	Df	Df.res	Pr(>F)
--	---	----	--------	--------

```
## Response      11.0923  1      95 0.001235 **
## ROI           4.5808  2      95 0.012612 *
## Response:ROI  0.9456  2      95 0.392065
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(Hipp_seg_Sham, ddf = "Kenward-Roger")
```

```
## Linear mixed model fit by REML. t-tests use Kenward-Roger's method [
## lmerModLmerTest]
## Formula: BOLD ~ Response * ROI + (1 | ID)
## Data: Hip_data_seg_enc
##
## REML criterion at convergence: -2.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.96834 -0.51460 -0.01914  0.46684  2.28799
##
## Random effects:
## Groups Name Variance Std.Dev.
## ID      (Intercept) 0.02178  0.1476
## Residual          0.03811  0.1952
## Number of obs: 120, groups: ID, 20
##
## Fixed effects:
##              Estimate Std. Error    df t value Pr(>|t|)
## (Intercept)      0.27478    0.05472 68.61933   5.021 3.88e-06 ***
## ResponseIncorrect -0.15971    0.06173 95.00000  -2.587  0.0112 *
## ROIMid            -0.13309    0.06173 95.00000  -2.156  0.0336 *
## ROIPost           -0.15113    0.06173 95.00000  -2.448  0.0162 *
## ResponseIncorrect:ROIMid 0.01309    0.08731 95.00000   0.150  0.8811
## ResponseIncorrect:ROIPost 0.10991    0.08731 95.00000   1.259  0.2112
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) RspnsI ROIMid ROIPost RI:ROIM
## RspnsInccrrc -0.564
## ROIMid        -0.564  0.500
## ROIPost       -0.564  0.500  0.500
## RspnsI:ROIM   0.399 -0.707 -0.707 -0.354
## RspnsI:ROIPost 0.399 -0.707 -0.354 -0.707  0.500
```

```
emmip(Hipp_seg_Sham, Response~ROI)
```



```
estimated_means <- get_means(Hipp_seg_Shram, "Response*ROI")
pander(estimated_means)
```

Response	ROI	Mean	SE	df	CI_lower	CI_higher
Correct	Ant	0.2748	0.05472	68.62	0.1656	0.384
Incorrect	Ant	0.1151	0.05472	68.62	0.005892	0.2243
Correct	Mid	0.1417	0.05472	68.62	0.0325	0.2509
Incorrect	Mid	-0.004927	0.05472	68.62	-0.1141	0.1043
Correct	Post	0.1236	0.05472	68.62	0.01447	0.2328
Incorrect	Post	0.07385	0.05472	68.62	-0.03533	0.183

```
Hipp_contrasts <- emmeans(Hipp_seg_Shram, specs = pairwise ~ Response|ROI, type = "response")
Hipp_contrasts
```

```
## $emmeans
## ROI = Ant:
##   Response    emmean      SE   df lower.CL upper.CL
##   Correct    0.27478 0.0547 68.6  0.16560   0.384
##   Incorrect  0.11507 0.0547 68.6  0.00589   0.224
##
## ROI = Mid:
##   Response    emmean      SE   df lower.CL upper.CL
##   Correct    0.14169 0.0547 68.6  0.03250   0.251
```

```
## Incorrect -0.00493 0.0547 68.6 -0.11411 0.104
##
## ROI = Post:
## Response      emmean      SE    df lower.CL upper.CL
## Correct      0.12365 0.0547 68.6  0.01447  0.233
## Incorrect    0.07385 0.0547 68.6 -0.03533  0.183
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## ROI = Ant:
## contrast      estimate      SE df t.ratio p.value
## Correct - Incorrect  0.1597 0.0617 95 2.587  0.0112
##
## ROI = Mid:
## contrast      estimate      SE df t.ratio p.value
## Correct - Incorrect  0.1466 0.0617 95 2.375  0.0196
##
## ROI = Post:
## contrast      estimate      SE df t.ratio p.value
## Correct - Incorrect  0.0498 0.0617 95 0.807  0.4219
##
## Degrees-of-freedom method: kenward-roger
```

```
## -- right hippocampus
```

```
Hip_data_seg_enc <- subset(ROIdata, StimType=="Sham" & (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & Response=="Correct")
Hipp_seg_Sham <- lmer(BOLD ~ Response*ROI + (1|ID), data = Hip_data_seg_enc)
Anova(Hipp_seg_Sham, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##              F Df Df.res Pr(>F)
## Response    1.3453  1    95 0.2490
## ROI          0.1263  2    95 0.8815
## Response:ROI 0.1735  2    95 0.8410
```

```
## Plot
```

```
Hip_data_seg_enc <- subset(ROIdata, StimType=="Sham" & (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & Response=="Correct")
g.Hip_data_seg_enc <- Hip_data_seg_enc %>% group_by(StimType, Response, ID) %>%
  summarise(BOLD = mean(BOLD))

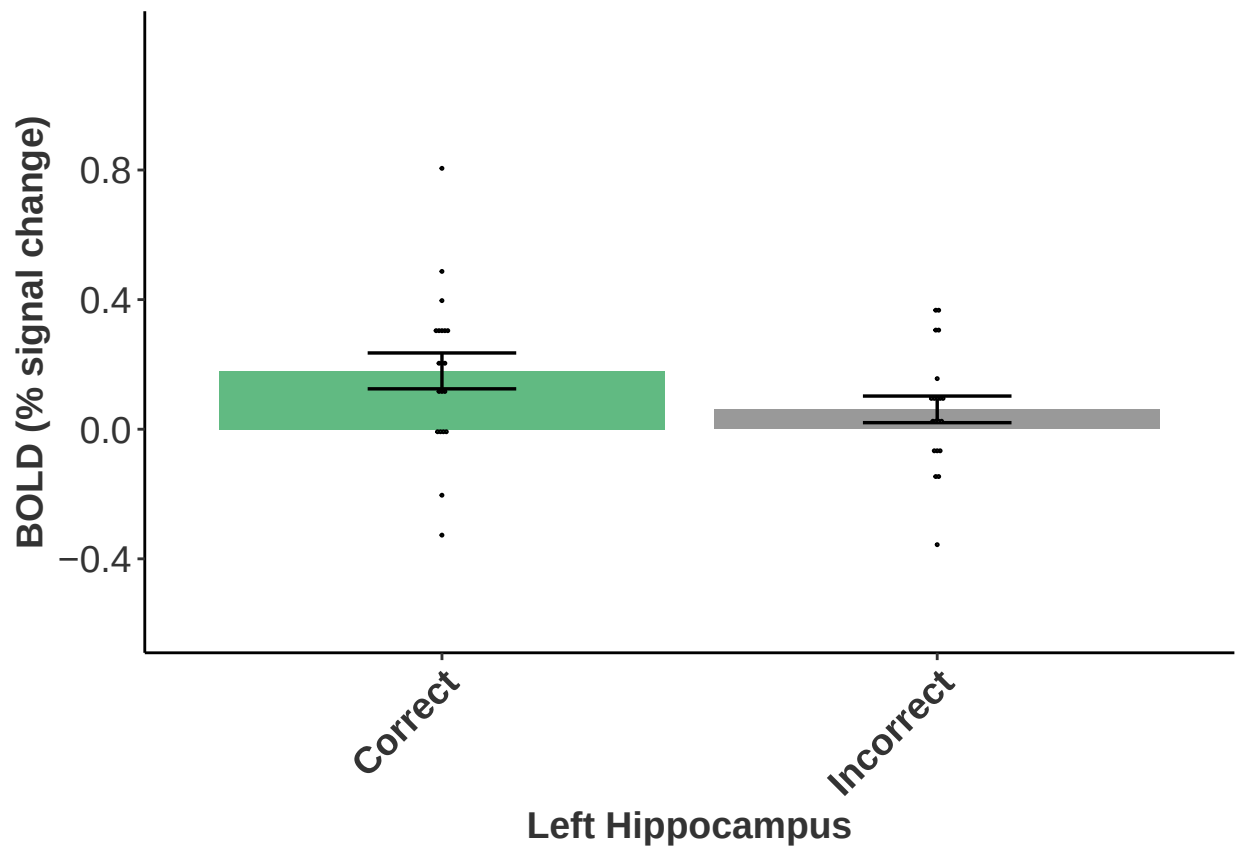
p.Hipp_seg_Sham <- ggplot(g.Hip_data_seg_enc, aes(x = Response, y = BOLD, fill = Response)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.2) +
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.6, 1.2)) +
```

```

theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
      axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold"),
      axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
      axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
      axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
      legend.text = element_blank(),
      legend.background = element_blank(),
      legend.key=element_blank(),
      legend.title=element_blank(),
      panel.background = element_blank()) +
scale_x_discrete(breaks=c("Correct", "Incorrect"),
labels=c("Correct","Incorrect")) +
scale_fill_manual(values=c("#61ba82", "#999999")) +
theme(panel.grid.major = element_blank(),
      panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
theme(strip.text.x = element_text(size = 14),
      strip.text.y = element_text(size = 14), legend.position = "none") + xlab("Left Hippocampus")

```

p.Hipp_seg_Sham



```

ggsave(p.Hipp_seg_Sham ,file="Figure_2H.tiff", width = 6, height = 9, dpi = 300, units = "cm", device =

```

Stim Conditions - segmented Left Hippocampus - Table S11 and Figure 20

```
## Left Hippocampus - TI 1:3 condition - Table S11.1
## LMM model comparing correct vs incorrect estimates
```

```
TI1_3_Hip_data_seg_enc <- subset(ROIdata, (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & Response!="C>I" & "
BA_seg <- lmer(BOLD ~ Response*ROI + (1|ID), data = TI1_3_Hip_data_seg_enc)
Anova(BA_seg, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##              F Df Df.res Pr(>F)
## Response    6.6161  1    95 0.01166 *
## ROI          0.7125  2    95 0.49302
## Response:ROI 1.4183  2    95 0.24722
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#emmip(BA_seg, ROI~Response)
estimated_means <- get_means(BA_seg, "Response*ROI")
pander(estimated_means)
```

Response	ROI	Mean	SE	df	CI_lower	CI_higher
Correct	Ant	0.1123	0.06791	47.59	-0.02424	0.2489
Incorrect	Ant	-0.06722	0.06791	47.59	-0.2038	0.06936
Correct	Mid	0.01991	0.06791	47.59	-0.1167	0.1565
Incorrect	Mid	-0.07153	0.06791	47.59	-0.2081	0.06505
Correct	Post	-0.01407	0.06791	47.59	-0.1506	0.1225
Incorrect	Post	-0.03693	0.06791	47.59	-0.1735	0.09965

```
## Left Hippocampus - TI 1:1 condition - Table S12.2
```

```
TI1_1_Hip_data_seg_enc <- subset(ROIdata, (ROI=="Ant" | ROI=="Mid" | ROI=="Post") & Response!="C>I" & "
BA_seg <- lmer(BOLD ~ Response*ROI + (1|ID), data = TI1_1_Hip_data_seg_enc)
Anova(BA_seg, test="F")
```

```
## Analysis of Deviance Table (Type II Wald F tests with Kenward-Roger df)
##
## Response: BOLD
##              F Df Df.res Pr(>F)
## Response    0.0273  1    95 0.86905
## ROI          4.7586  2    95 0.01073 *
## Response:ROI 0.3992  2    95 0.67196
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#emmip(BA_seg, ROI~Response)
estimated_means <- get_means(BA_seg, "Response*ROI")
pander(estimated_means)
```

Response	ROI	Mean	SE	df	CI_lower	CI_higher
Correct	Ant	0.2076	0.05535	63.78	0.09704	0.3182
Incorrect	Ant	0.1653	0.05535	63.78	0.05469	0.2758
Correct	Mid	0.09211	0.05535	63.78	-0.01847	0.2027
Incorrect	Mid	0.08282	0.05535	63.78	-0.02775	0.1934
Correct	Post	0.0434	0.05535	63.78	-0.06718	0.154
Incorrect	Post	0.07763	0.05535	63.78	-0.03294	0.1882

```
# Plotting average for simpler figure
```

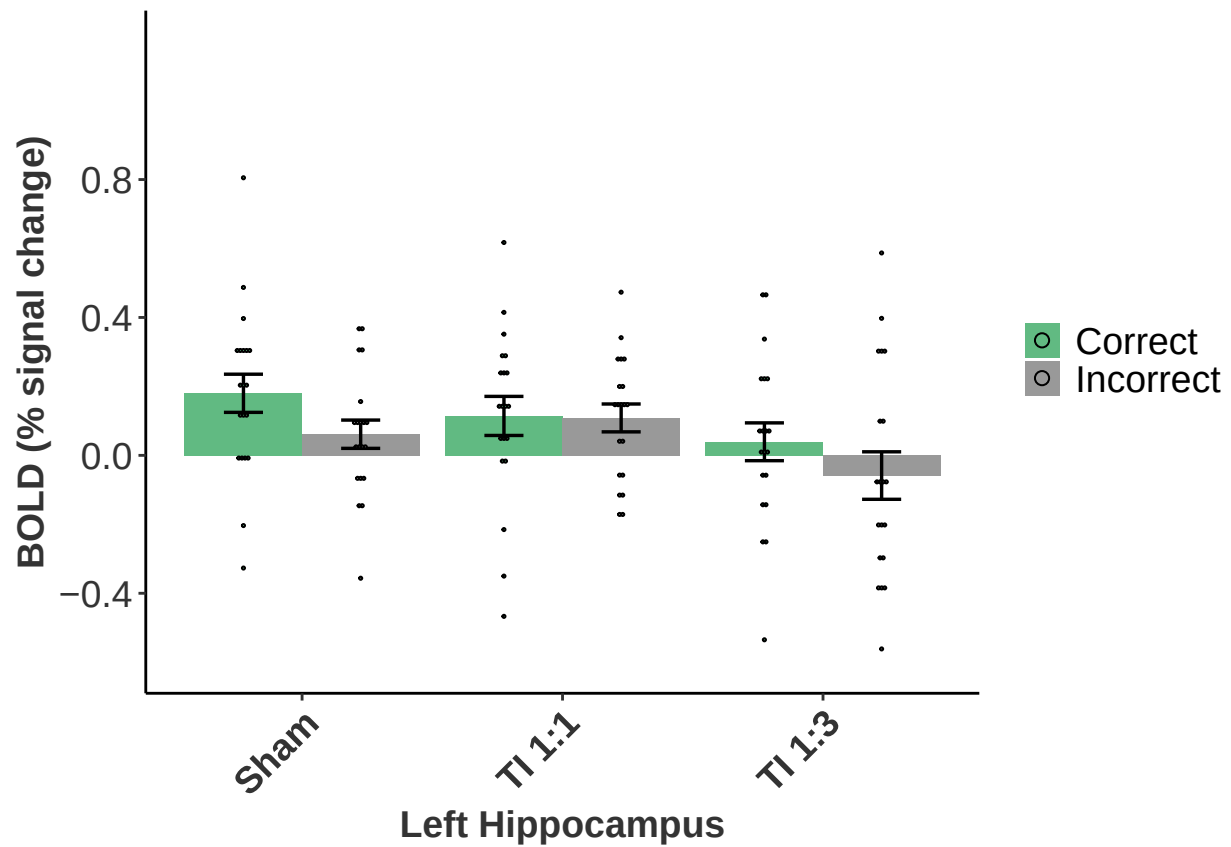
```
Hip_data_enc <- subset(ROIdata, Response!="C>I" & TaskStage=="Encode" & hemisphere=="L" & ROI!="Hippocampus")
g.Hip_data_enc <- Hip_data_enc %>% group_by(StimType, Response, ID) %>%
  summarise(BOLD = mean(BOLD))
```

```
## 'summarise()' has grouped output by 'StimType', 'Response'. You can override
## using the '.groups' argument.
```

```
p.Hipp <- ggplot(g.Hip_data_enc, aes(x = StimType, y = BOLD, fill = Response)) +
  stat_summary(fun = mean, geom = "bar", position = position_dodge(width = 0.9)) +
  stat_summary(geom = "errorbar", fun.data = mean_se, position = position_dodge(0.9), size = 0.6, width = 0.2) +
  scale_y_continuous(name = "BOLD (% signal change)",
    breaks = seq(-.8, 1, .4),
    limits=c(-.6, 1.2)) +
  theme(plot.title = element_text(color="black", size=14, face="bold", hjust = 0.5),
    axis.text.x = element_text(color = "grey20", size = 14, angle = 45, face = "bold", hjust = 1),
    axis.text.y = element_text(color = "grey20", size = 14, angle = 0, face = "plain"),
    axis.title.x = element_text(color = "grey20", size = 14, angle = 0, face = "bold"),
    axis.title.y = element_text(color = "grey20", size = 14, angle = 90, face = "bold"),
    legend.text = element_blank(),
    legend.background = element_blank(),
    legend.key=element_blank(),
    legend.title=element_blank(),
    panel.background = element_blank()) +
  scale_fill_manual(values=c("#61ba82", "#999999")) +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(), axis.line = element_line(colour = "black")) +
  theme(strip.text.x = element_text(size = 14),
    strip.text.y = element_text(size = 14), legend.text = element_text(size=14),
    legend.key.size = unit(0.5, 'cm')) + xlab("Left Hippocampus")

p.Hipp
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with 'binwidth'.
```



```
ggsave(p.Hipp ,file="Figure_20.tiff", width = 10, height = 8, dpi = 300, units = "cm", device = "tiff")
```

```
## Bin width defaults to 1/30 of the range of the data. Pick better value with 'binwidth'.
```

```
#
```